Chapter 12. Planar Graphs

Do Not Cross

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12.1 Euler's theorem for planar graphs

Motivation. Euler's formula $V + F = E + 2$ plus edge counts around faces rules out $K_{3,3}$ and K_5.

Definition 12.1 (Planar graph)

A graph is **planar** if it can be drawn in the plane with no two edges crossing. The edges of such a drawing cut the plane into regions, the **faces**; one of them, the outer face, is unbounded.

Theorem 12.2 (Euler's theorem on planar graphs)

A connected planar graph with V vertices, E edges and F faces satisfies $V + F = E + 2$.

Proof. Induction on E. $E = 1$: one edge ($V = 2$, $F = 1$) or one loop ($V = 1$, $F = 2$). Let $E \geq 2$.

Step 1 (an edge on a cycle). If $G' = G - e$ is connected, e lies on a cycle, so the two faces beside e merge: G' has V, $E - 1$, $F - 1$, and $V + F - 1 = E - 1 + 2$ by induction.

Step 2 (a tree). Otherwise G is a tree: $V = E + 1$ (Theorem 10.4), $F = 1$. ■

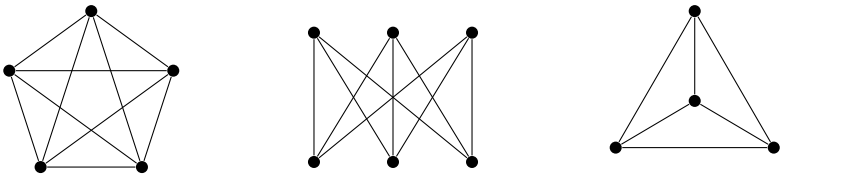
Example 12.3

$K_{3,3}$ is not planar, so the three houses, three wells problem has no solution.
Solution. $V = 6$, $E = 9$ would give $F = 5$. $K_{3,3}$ is simple bipartite, so each face has ≥ 4 edges and $4F \leq 2E$: $20 \leq 18$, a contradiction.

Hence $K_6 \supseteq K_{3,3}$ is not planar; K_4 is.

Example 12.4

K_5 is not planar.
Solution. $V = 5$, $E = 10$ give $F = 7$, and $3F \leq 2E$: $21 \leq 20$, a contradiction.



K_5 $K_{3,3}$ K_4, drawn without crossings

Figure. *The two minimal non-planar graphs and a plane drawing of K_4.*

Subdividing an edge or suppressing a degree-2 vertex preserves planarity; graphs related by these moves are **edge-equivalent** (homeomorphic).

Theorem 12.5 (Kuratowski)

A graph is not planar iff it contains a subgraph that is edge-equivalent to K_5 or $K_{3,3}$.

Proof. "If": such a subgraph is non-planar (Examples 12.3, 12.4), hence so is the graph. The book states "only if" without proof. ■

12.2 Polyhedra

Motivation. Euler's formula leaves room for only five regular polyhedra.

For a convex polyhedron (its 1-skeleton), faces have ≥ 3 edges and vertices degree ≥ 3. It is **regular** if every face is an ℓ-cycle and every vertex has degree d.

Theorem 12.6

A convex polyhedron with V vertices, F faces and E edges satisfies $V + F = E + 2$.

Proof (angle counting). Project P onto a plane perpendicular to no face. Faces project to convex polygons; the outer boundary B is a convex v_1-gon, and $v_2 = V - v_1$ vertices project inside. Each interior point is covered by one front and one back face. Let S be the sum of all angles of projected faces.

Step 1 (by vertices). An inside vertex contributes $360°$; on B, front and back faces each fill the interior angle, contributing twice $(v_1 - 2)180°$:

$$S = (v_1 - 2)360 + 360v_2 = (V - 2)360. \tag{12.1}$$

Step 2 (by faces). With f_i the edge numbers of the faces,

$$\sum_{i=1}^{F} f_i = 2E, \tag{12.2}$$

and a convex k-gon has angle sum $(k - 2)180°$, so

$$S = \sum_{i=1}^{F}(f_i - 2)180 = 180\sum_{i=1}^{F} f_i - 360F = 360(E - F). \tag{12.3}$$

(12.1) and (12.3) give $V - 2 = E - F$. ■

Corollary 12.7

In a convex polyhedron, $3F \leq 2E$.

Proof. $f_i \geq 3$ in (12.2). ■

Proposition 12.8

In a convex polyhedron, $3V \leq 2E$.

Proof. With vertex degrees $c_i \geq 3$,

$$\sum_{i=1}^{V} c_i = 2E. \tag{12.4}$$

■

Lemma 12.9

In a convex polyhedron, $E \leq 3V - 6$ and $E \leq 3F - 6$.

Proof. Corollary 12.7: $E + 2 = V + F \leq V + \frac{2E}{3}$, so $E \leq 3V - 6$. Proposition 12.8: $E + 2 \leq F + \frac{2E}{3}$, so $E \leq 3F - 6$. ■

Remark

Remark[$E \le 3V-6$ for every simple planar graph [Supplement]] For simple connected planar G with $V \ge 3$ (add edges between components): a tree has $E = V-1 \le 3V-6$; otherwise faces have closed boundary walks of length ≥ 3, so $3F \le 2E$ and Euler gives $E \le 3V-6$. Hence some vertex has degree ≤ 5 (else $2E \ge 6V$).

Lemma 12.10

Every convex polyhedron has a face with at most five edges.

Proof. By (12.2) and Lemma 12.9,

$$\sum_{i=1}^{F} f_i = 2E \le 6F - 12 < 6F. \tag{12.5}$$

■

Lemma 12.11

Every convex polyhedron has a vertex of degree at most five.

Proof. By (12.4) and Lemma 12.9,

$$\sum_{i=1}^{V} c_i = 2E \le 6V - 12 < 6V. \tag{12.6}$$

■

Put a vertex in each face of a plane G and join the two faces beside each edge e by an edge crossing e: G^* is planar with F vertices, E edges, V faces.

Definition 12.12 (Dual graph)

G^* is the **dual graph** of the plane graph G.

The dual of a regular polyhedron of type (d, ℓ) is regular of type (ℓ, d).

Classification. Lemmas 12.10, 12.11 give $d, \ell \in \{3,4,5\}$, and by (12.2)

$$E = \frac{F\ell}{2}. \tag{12.7}$$

(A) $d=3$: $3V = 2E$, $3F = E+6$, $(6-\ell)F = 12$; $\ell = 3,4,5$ give $(V,E,F) = (4,6,4)$ (tetrahedron), $(8,12,6)$ (cube), $(20,30,12)$ (dodecahedron).

(B) $d=4$: $E = 2F-4$, $(4-\ell)F = 8$; $\ell = 3$: $(6,12,8)$, octahedron.

(C) $d=5$: $3E = 5F-10$, $(10-3\ell)F = 20$; $\ell = 3$: $(12,30,20)$, icosahedron.

Theorem 12.13

There are exactly five regular polyhedra: the tetrahedron, the cube, the dodecahedron, the octahedron and the icosahedron.

Proof. (A)–(C) leave exactly these five parameter sets, each realised by the named solid. ■

	d	ℓ	V	E	F
Tetrahedron	3	3	4	6	4
Cube	3	4	8	12	6
Octahedron	4	3	6	12	8
Dodecahedron	3	5	20	30	12
Icosahedron	5	3	12	30	20

12.3 Colouring maps

Motivation. Map colouring is vertex colouring of the dual; Kempe chains give five colours.

Colouring faces so that neighbours differ is properly colouring G^*. K_4 shows four colours are needed.

Proposition 12.14

Every planar graph can be properly vertex-coloured with six colours.

Proof. Induction on V, G simple. By the Remark after Lemma 12.9 some A has degree ≤ 5; colour $G - A$ by induction and give A a colour missing from its neighbours. ■

Theorem 12.15 (Five colour theorem)

Every planar graph can be properly vertex-coloured with five colours.

Proof. Induction on V, G simple. Take $\deg A \leq 5$ and colour $G' = G - A$. If A's neighbours use ≤ 4 colours, done. Otherwise they are $y_1, \ldots, y_5$ clockwise, coloured $1, \ldots, 5$.

Step 1 (Kempe chains). Let K be the component of y_1 in the subgraph induced by colours $1, 3$. Swapping $1, 3$ on K keeps the colouring proper; if $y_3 \notin K$, colour 1 becomes free at A.

Step 2 (both chains cannot exist). If $y_3 \in K$, a 1–3 path $y_1 \to y_3$ and A form a cycle C separating y_2 from y_4. Likewise we are done unless a 2–4 path $y_2 \to y_4$ exists; it would meet $C - A$ (coloured $1, 3$) at a vertex, impossible. ■

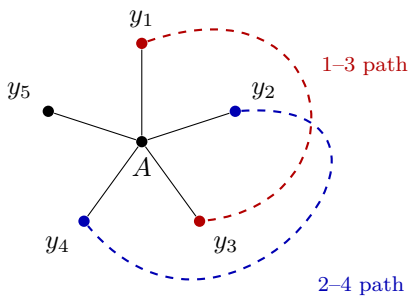


Figure. *Kempe chains: a 1–3 path $y_1 \to y_3$ and a 2–4 path $y_2 \to y_4$ would cross at a vertex.*

Remark

Remark[The four colour theorem] Four colours suffice: Appel–Haken (1976, computer check of 1936 configurations; 633 by Robertson–Sanders–Seymour–Thomas, 1996); machine-verified

by Gonthier–Werner (2005).

12.4 Selected exercises

Motivation. Euler's formula plus two double counts gives one linear relation.

The *length* of a face counts boundary edges (twice if the face lies on both sides); face lengths and degrees each sum to $2E$.

Exercise 12.15 (Polyhedra without triangles)

Let P be a convex polyhedron with no triangular faces. Prove that $E \le 2V - 4$.

Solution.

Step 1 (faces). $4F \le 2E$, so $F \le \frac{E}{2}$.

Step 2 (Euler). By Theorem 12.6, $E + 2 = V + F \le V + \frac{E}{2}$, so $E \le 2V - 4$.

Equality iff all faces are quadrilaterals (cube: $V = 8$, $E = 12$). ■

Exercise 12.19 (Trivalent (a, b)-faced polyhedra)

Let P be a convex polyhedron whose faces are all a-gons or b-gons and in which every vertex lies on exactly three edges. Let p_a, p_b be the numbers of a-gonal and b-gonal faces and n the number of vertices.
(a) Express the number E of edges in two ways.
(b) Prove that $p_a(6 - a) + p_b(6 - b) = 12$.

Solution.

(a) $2E = a\,p_a + b\,p_b$ (faces) and $2E = 3n$ (vertices).

(b) Multiply $n + p_a + p_b = E + 2$ (Theorem 12.6) by 6 and use $6n = 4E$:

$$4E + 6p_a + 6p_b = 6E + 12, \qquad \text{so} \qquad 6p_a + 6p_b - 2E = 12,$$

and substitute $2E = a\,p_a + b\,p_b$.

E.g. a fullerene (pentagons and hexagons) has exactly 12 pentagons. ■

Exercise 12.22 (Octahedrites)

Let G be a connected planar graph in which each face has length 2, 3 or 4, with p_2, p_3, p_4 faces of these lengths. Assume that every vertex has degree 4 and that $p_2 + p_3 = 8$. Prove that $p_2 = 0$ and $p_3 = 8$.

Solution.

Step 1 (two counts). $V = \frac{E}{2}$, $2E = 2p_2 + 3p_3 + 4p_4$, $F = p_2 + p_3 + p_4$.

Step 2 (Euler). Theorem 12.2 gives $F = \frac{E}{2} + 2$; times 4:

$$4p_2 + 4p_3 + 4p_4 = 2E + 8 = 2p_2 + 3p_3 + 4p_4 + 8, \qquad \text{so} \qquad 2p_2 + p_3 = 8.$$

Step 3 (conclusion). Subtracting $p_2 + p_3 = 8$: $p_2 = 0$, $p_3 = 8$. ■